

2. Real-Time Object Detection and Counting from Video Streams

Abstract: Many companies use computer vision to detect, count, and track objects in video streams. Examples include traffic monitoring, warehouse automation, retail analytics, safety monitoring, and event analysis. In this project, I will build a real-time object detection system that can process videos and count objects of interest.

The project will use a pretrained object detection model such as YOLOv8, YOLOv5, or a lightweight MobileNet-based detector.⁵ Using a pretrained model is practical because it reduces the amount of training needed and makes the project possible with Kaggle GPU or even CPU inference. The model will be fine-tuned on a small custom dataset or used directly on a public dataset such as COCO-style images.⁶

The system will take a video file or webcam-like input, detect objects frame by frame, draw bounding boxes, and count selected object classes. I will also measure important engineering metrics such as inference latency, frames per second, model size, and memory usage. These metrics are important because real machine learning systems are judged not only by accuracy, but also by speed, cost, and reliability.

The project will include a simple dashboard where users can upload a short video and see the detections, counts, and performance metrics. I will also compare different model sizes, such as nano, small, and medium versions, to show the trade-off between speed and accuracy.

The final deliverable will be a GitHub repository with clean code, a demo video, a short technical report, and deployment instructions. This project will prepare me for AI/ML engineering roles because it shows practical computer vision, real-time inference, benchmarking, and deployment.

⁵<https://docs.ultralytics.com/>

⁶<https://cocodataset.org/>